QUESTION – Differentiate between JPA, Hibernate and Spring JPA

1. JPA (Java Persistence API)

JPA is a java specification for storing, retrieving, updating, and deleting data from a database using java objects. It is not a framework and does not provide its own implementation. It only defines how ORM (Object Relational Mapping) should work.

2. Hibernate

Hibernate is an ORM framework and the most popular implementation of JPA. It follows all the rules defined by JPA and also provides many additional features such as caching, lazy loading and better performance optimization.

3. Spring Data JPA (Spring JPA)

Spring Data JPA is a Spring Framework module built on top of JPA. It reduces boilerplate code by providing ready-made repository interfaces like JPA Repository. It internally uses a JPA implementation to communicate with the database.

Difference between JPA, Hibernate and Spring Data JPA

Feature	JPA	Hibernate	Spring Data JPA
Definition	A Java specification for ORM.	An ORM framework that implements JPA.	A Spring module that simplifies JPA usage.
Type	Specification	Framework	Spring Framework module
Implementation	Does not implement anything itself.	Provides complete implementation of JPA.	Uses JPA (mostly Hibernate) internally.
Purpose	Defines standard rules for database operations.	Performs actual database operations.	Reduces coding effort by providing ready-made repositories.
Dependency	Needs implementation like Hibernate.	Can work independently or with JPA.	Depends on both JPA and a JPA provider like Hibernate.

Feature	JPA	Hibernate	Spring Data JPA
CRUD Operations	Must be written manually using Entity Manager.	Uses Session or Entity Manager.	Uses JPA Repository; most CRUD methods are already available.
Boilerplate Code	More code	Moderate code	Very little code
Ease of Use	Moderate	Easy	Easiest
Extra Features	Only defines standards.	Caching, lazy loading, query optimization, etc.	Pagination, sorting, derived query methods, custom repositories.